

DINO for Anomaly Detection on Battery Cells

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Abstract

Battery cell anomaly detection in industrial setups is challenging due to severe class imbalance and the small spatial footprint of defects. To address this, we evaluate the DINOv3 vision transformer on a dataset of 336 images across three paradigms: parameter-efficient fine-tuning (PEFT), zero-shot classification, and one-class representation learning. Results show the DINOv3 Adapter outperforms full fine-tuning and zero-shot baselines in supervised classification. In unsupervised settings, Isolation Forest effectively separates normal and abnormal cells without defect supervision. Additionally, for zero-shot tasks, localization via Grounding DINO proved more robust than global similarity approaches. Furthermore, our proposed DINOv3-YOLO architecture matches native YOLO detection performance while drastically reducing trainable parameters, proving DINOv3 to be a highly effective, parameter-efficient backbone for low-data industrial inspection. Code implementation is available at [github](#).

1. Methodology and Material

1.1. Dataset and preprocess

The dataset has been gathered and manually annotated from the Fraunhofer Innovation Platform at the University of Twente. The dataset has been annotated for a detection task, while 3 different labels were used, including battery cell, text, and abnormality. The real distribution of anomalies in the battery cells is highly imbalanced, and only a few of the new battery cells have anomalies; therefore, the gathered dataset focuses on second-hand battery cells that have a higher chance of being defective. Each battery cell in this dataset has 6 images, which 4 of them are from different sides of the cell, and 2 of them are from top view and bottom view. This specification of data required a cautious, cell-wise data split to prevent correlation between the train set and validation set.

Figure 1 shows the image-wise distribution of dataset. The entire dataset has 336 images that has been captured from 56 unique battery cells. Due to the low number of

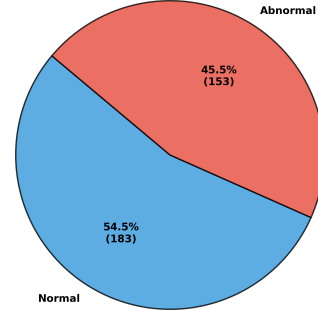


Figure 1. Image-wise distribution of data.

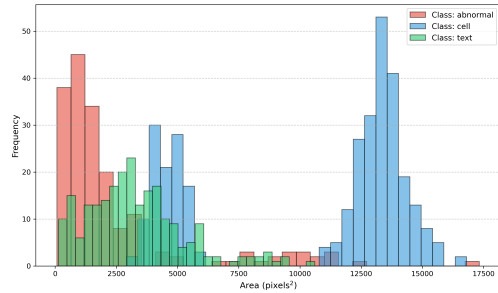


Figure 2. The area distribution of bounding boxes

samples in the dataset, a 5-fold-cross-validation approach has been pursued to reduce the possible effect of split distribution on results.

Figure 2 illustrates the distribution of the area of bounding boxes per class, which shows that the abnormality comprises a minority of pixels in the image.

1.2. Preliminary

The proposed architecture consists of the ViT-Based DINOv3 [7] model as the backbone for vision feature extraction. Moreover, to enhance the performance of the architecture for downstream tasks, the backbone model should be fine-tuned. For this purpose, Parameter-efficient fine-tuning (PEFT) methodologies are used as modern techniques to adapt large-scale foundation models to downstream tasks while mitigating severe computational and storage over-

head. To establish a comparative baseline, the fine-tuning of top layers (top tuning) [1] is frequently applied. Alternatively, Low-Rank Adaptation (LoRA) [2] is implemented by freezing the original pre-trained network weights and exclusively optimizing injected low-rank decomposition matrices, thereby enabling practical and efficient model adaptation. Furthermore, prompt tuning strategies specifically engineered for vision-language models, such as Context Optimization (CoOp) [8], are deployed by modeling prompt context words as continuous, learnable vectors, successfully aligning fixed model representations with new visual categories without updating the underlying image or text encoders. In this study, PEFT methods include top-tuning, LoRA, Adapters, and Vision Prompt Tuning (VPT).

1.3. Classification

1.3.1. Supervised Method

In this approach, the CLS token from the DINOv3 ViT-b16 model has been used, and a simple Multi-Layer Perceptron (MLP) has been added on top of it for binary classification between normal and abnormal images.

1.3.2. Zero-shot Anomaly Detection

To evaluate the feasibility of classification of abnormalities in battery cells without any task-specific training, two zero-shot vision-language approaches were benchmarked on the same 5-fold cross-validated dataset, with each fold split into abnormality (defective) and normal classes.

TinyCLIP

The first method is TinyCLIP, a compressed variant of CLIP pretrained on the LAION-400M dataset, accessed via Hugging Face’s CLIPModel/CLIPProcessor. Classification was performed using standard CLIP zero-shot inference: each validation image was encoded into a normalized image embedding, and each candidate class label was encoded into a normalized text embedding. Cosine similarity (scaled by a temperature factor of 100) between the image embedding and each text embedding was computed, and the class with the highest similarity score was taken as the prediction. Since CLIP-style models are sensitive to how prompts are phrased, six distinct prompt-engineering strategies were tested to determine which framing best captured battery defects:

- Basic: simple, direct labels (“a photo of a defective/perfect battery cell”)
- Physical Damage: focused on mechanical defects (scratches, dents, cracks)
- Chemical Defects: focused on corrosion, rust, and leakage
- Deformation: focused on swelling and structural failure
- QA Inspection: framed the task in an industrial quality-control context

- Comprehensive (ensemble): combined multiple defect types into a single descriptive prompt

Each strategy used one positive (defective) and one negative (normal) text prompt as the two-class decision boundary, evaluated independently across all 5 folds.

Grounding DINO

The second method used Grounding DINO, an open-vocabulary object detector that localizes regions matching free-text phrases rather than performing whole-image classification. A single compound text prompt listing defect-related terms (“scratch. dent. rust. crack. leak. stain. bulge. dirty spot. broken surface. damaged metal.”) was used to query each image. Detection was run with a box confidence threshold of 0.35 and a text threshold of 0.25. An image was classified as abnormal if at least one bounding box matching the defect vocabulary was returned above threshold, and normal otherwise, effectively converting the detector’s localization output into a binary presence/absence classifier.

1.3.3. Representation Learning

We investigate a unsupervised direction: treating the frozen DINOv3 embedding space directly as an anomaly score, without ever exposing the model to a labeled abnormal example, motivated by deployment scenarios where abnormal samples are scarce or unknown in advance.

The methodology includes extracting a 768-dimensional CLS-token embedding per image from a frozen DINOv3 backbone (dinov3-vitb16-pretrain-lvd1689m) and fit three classical one-class models exclusively on the 135 normal training embeddings: Isolation Forest [4], One-Class SVM [6], and a Mahalanobis distance model.

1.4. Detection

To develop an architecture for the detection task, the YOLOv26 neck and head layers have been merged into the DINOv3 backbone, and the code has been modified to become compatible with the Ultralytics pipeline; therefore, the architecture has been prepared for training detection models with the same pipeline that YOLO models are trained with. The proposed architecture for the combination of DINO and YOLO is shown in Figure 3. Since YOLOv26 uses a Feature Pyramid Network (FPN), the simple connection between the latest layers on DINOv3 and YOLOv26 is not possible. To address this problem, ViTDET [3] provides a solution to create an FPN architecture in a transformer-based model. Based on ViTDET, the intermediate layer is designed to get vision tokens from the latest layer of DINOv3 and prepare them through the FPN network to be send into the YOLOv26 neck.

2. Results

Table 1 shows results across the six TinyCLIP prompt variants; performance varied substantially, from near chance

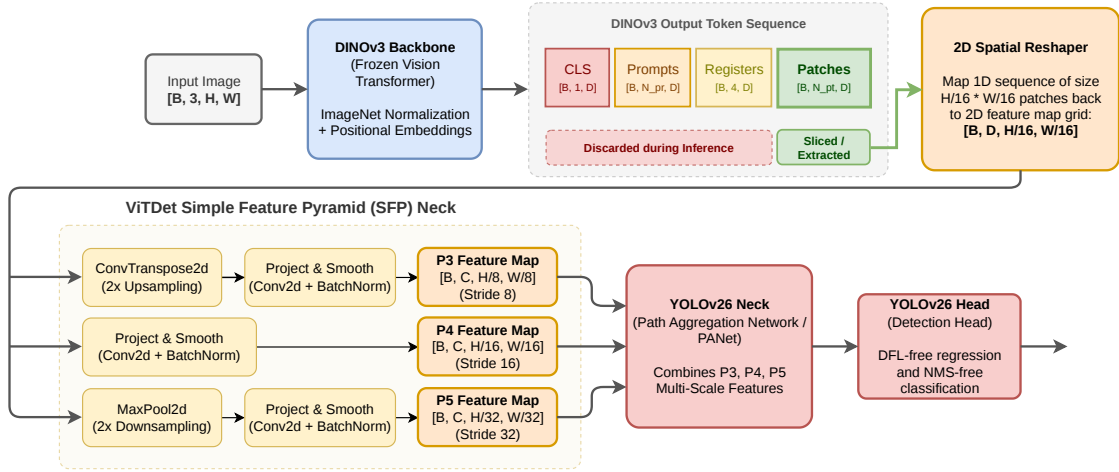


Figure 3. The architecture to combine YOLOv26 and DINOv3 together.

level (28.87% accuracy for the “physical damage” prompt) up to 66.07% for the “chemical defects” prompt, which consistently performed best across all folds and all metrics. This spread highlights the sensitivity of CLIP-style contrastive classification to prompt wording: framings tied to concrete, visually distinctive cues (rust, corrosion, leakage) outperformed broader or more abstract framings (e.g., generic “QA inspection” language), even though all prompts targeted the same underlying defect categories. Moreover, Grounding DINO used a single compound detection prompt rather than a tuned pair of class descriptions and achieved the strongest overall performance of any zero-shot configuration tested (69.64% accuracy, 0.693 macro F1), outperforming even TinyCLIP’s best prompt strategy by roughly 3.5 points of accuracy and 0.035 F1. This suggests that reframing the classification problem as open-vocabulary localization (does a defect-matching region exist anywhere in the image?) is more robust than whole-image embedding similarity for this task, likely because battery defects (scratches, rust spots, dents) are typically small, localized surface features rather than global appearance changes, a pattern that plays to a detector’s strengths and is harder for a single global CLIP embedding to capture.

Table 1. Zero-shot performance compatisation

Strategy / Method	Accuracy	Precision	Recall	F1-score
Grounding DINO	69.64%	0.7289	0.7111	0.6933
Basic	47.32%	0.5077	0.5041	0.4209
Physical Damage	28.87%	0.2910	0.2929	0.2884
Chemical Defects	66.07%	0.6860	0.6741	0.6580
Deformation	58.93%	0.5862	0.5704	0.5590
QA Inspection	46.13%	0.5285	0.5033	0.3422
Comprehensive (ensemble)	56.55%	0.6319	0.5256	0.4252

Fold-level results for both methods were reasonably stable (no fold was a dramatic outlier), suggesting these performance differences reflect genuine differences in the two paradigms rather than noise from the fold splits.

Table 2 reports performance on the val set (48 normal, 33 abnormal). Isolation Forest is the strongest model (F1=0.727, AUROC=0.829). Mahalanobis collapses under the high embedding dimensionality – estimating a 768×768 covariance matrix from 135 samples is severely underdetermined – flagging nearly all images as abnormal (recall=1.0, precision=0.41).

Model	Accuracy	Precision	Recall	F1
Isolation Forest	0.815	0.909	0.606	0.727
One-Class SVM	0.654	0.553	0.788	0.650
Mahalanobis distance	0.407	0.407	1.000	0.579

Isolation Forest across 5 seeds (ratio≈1.0):
F1: 0.711 ± 0.035 AUROC: 0.849 ± 0.033 Recall: 0.576 ± 0.044

Table 2. One-class anomaly detection results on the val set.

Regarding reliability, across three disjoint test sets (abnormal:normal ratios 0.83–1.25, 5 seeds each), F1 and AUROC remain stable (F1: 0.667–0.718, AUROC: 0.848–0.861), confirming the results are not an artifact of a favorable split. McNemar’s test [5] confirms Isolation Forest significantly outperforms both One-Class SVM ($p = 0.016$) and Mahalanobis ($p = 0.031$), while SVM and Mahalanobis are statistically indistinguishable ($p = 1.0$). These results suggest DINOv3’s pretrained representations carry enough structure for competitive anomaly detection without any abnormal supervision.

Furthermore, PEFT methods are used across the DINOv3 model for both supervised training of the classifica-

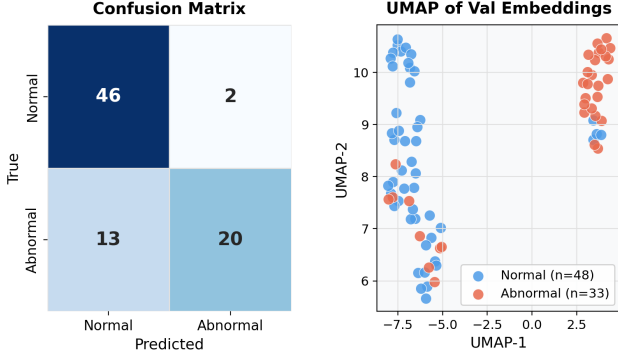


Figure 4. Confusion matrix and UMAP projection of DINOv3 val-set embeddings under the Isolation Forest one-class model. Normal and abnormal cells form largely separable clusters in the frozen feature space despite no abnormal supervision during training.

tion task and the detection task. The PEFT methods include top-tuning, LoRA, Adapter, and VPT. Table 3 shows the related hyperparameters and the best options for each task.

Table 3. PEFT Configurations hyperparameters

PEFT Method & Sweep Parameters	Best Config (Class.)	Best Config (Detect.)
Adapter Bottleneck Dimension $d \in \{32, 64\}$ Targeted Blocks $b \in \{\text{last 2, last 4}\}$	$d = 64$ $b = \text{last 4}$	$d = 32$ $b = \text{last 2}$
LoRA Rank $r \in \{8, 16\}$ Targeted Blocks $b \in \{\text{last 2, last 4}\}$	$r = 8$ $b = \text{last 2}$	$r = 8$ $b = \text{last 4}$
Vis. Prompt Prompt Length $t = 10$ Mode $\in \{\text{shallow, deep (last 2), deep (last 4)}\}$	Mode = deep $b = \text{last 4}$	Mode = shallow
Top Tuning N/A (Task-specific heads only)	Head only	Heads only

Table 4 shows the comprehensive comparison of the classification results with each other while DINOv3 with adapters has the best performance for binary classification of anomaly cells. the result of DINO with adapters are significantly better than the result of top tuning, which shows how PEFT methods enhance the performance while keeping the number of trainables low. Moreover, with the comparison of zero-shot models with the baseline, it becomes clear that the zero-shot models reduce the uncertainty a little, but the outcome is not significant, so they still cannot replace the supervised method in real applications. Another important observation is the close performance of the top tuning method and the representation learning method. This shows that the extracted features from DINOv3 are informative enough so that by only considering healthy samples, we can reach the same result as a normal supervised method with all of the labels.

Table 5 shows the detection model results. In this section, YOLO26 nano and small are baselines to compare

the DINO-YOLO architecture with them. The presented table has both detection results and classification metrics that were extracted from the detection task. In this comparison, the DINO-YOLO model reached close performance and even better performance in some metrics which indicates the DINO-YOLO is performing properly. The other observation indicates that extracted classification metrics are close to the best classification result, but still, the classification model is superior for the classification task.

3. Discussion and Conclusion

Anomaly detection in industrial setups is a growing field, and the emergence of AI improves it significantly. In this research, the performance of foundation model-based methods is evaluated to check the feasibility of these approaches to address challenges in this area.

Data imbalance, scarce abnormal cases, and the efficiency of manual annotations are among the main challenges for anomaly detection in industrial setups, while using modern methods in AI can enhance the bottleneck in this challenges. For this purpose, first, a zero-shot approach is used that mostly depends on prompt engineering and the knowledge of an expert about the anomaly itself. Since this approach is not dependent on supervised training, the aim is to balance the dataset, and manual annotation does not affect its performance, so it is a good candidate for rapid development of anomaly detectors. Although the concept of using zero-shot models seems promising, the performance of models on battery cell anomaly detection is not good enough, and it did not improve the baseline detector much; therefore, this approach cannot be used in real setups.

The other method that could be used for anomaly detection is to use the representation learning concept by relying on the rich features that a foundation model can extract from the image, and then identify the normal variant of features for normal cells. This method usually fits a machine learning-based (ML) algorithm on the feature maps during the training phase, while during the inference phase, it uses the same ML model to see if the input features follow the distribution of healthy images in the training set or not. This method addresses the data imbalance and manual annotation, while the experiments show that the inference results are close to those of the supervised method that uses the DINO model with PEFTs. The main bottleneck for this approach is the dataset size; therefore, if the number of samples increases, it is expected that the performance of this model will improve, but due to imbalance characteristic of the data, increasing the size of the dataset will not necessarily enhance the supervised method.

Lastly, the use of the DINO model and PEFT methods results in the best performance for both binary classification and detection. This experiment shows that for limited data, the best performance comes from a supervised

Table 4. Classification Results

Model	PEFT	Total P.	Train P. %	F1 (%)	Rec. (%)	Prec. (%)	Acc. (%)
DINOv3	Adapter	22.2M	0.90% (199.2K)	90.07 $\pm$ 5.07	89.64 $\pm$ 4.83	90.61 $\pm$ 5.93	91.13 $\pm$ 4.67
DINOv3	LoRA	22.1M	0.11% (25.3K)	88.08 $\pm$ 5.49	86.81 $\pm$ 5.39	89.69 $\pm$ 7.58	89.57 $\pm$ 4.29
DINOv3	Vis. Prompt	22.1M	0.07% (16.1K)	86.20 $\pm$ 3.61	86.40 $\pm$ 3.99	86.33 $\pm$ 6.14	87.62 $\pm$ 2.94
DINOv3	Top Tuning	22.1M	0.00% (770)	75.23 $\pm$ 4.83	74.28 $\pm$ 7.46	77.27 $\pm$ 7.64	77.62 $\pm$ 5.89
TINYCLIP	None	19M	None	66	67	69	66
GROUNDING DINO	None	172M	None	69	71	73	70
Rep. Learning*	None	22.2M	None	73	60	91	81
Baseline	Always Normal	0	None	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	54.47 $\pm$ 5.83
Baseline	Always Abnormal	0	None	62.35 $\pm$ 5.39	100.00 $\pm$ 0.00	45.53 $\pm$ 5.83	45.53 $\pm$ 5.83

Table 5. Object Detection Models

Model	PEFT Method	Total P.	Train. P. / %	Bbox mAP50	Bbox IoU	Bbox DICE	Class F1 (%)	Class Rec (%)	Class Prec (%)	Class Acc (%)
YOLO26S	Top Tuning	19.3M	19.30M / 100%	70.70 $\pm$ 3.81	81.75 $\pm$ 1.97	89.54 $\pm$ 1.20	86.70 $\pm$ 4.29	80.18 $\pm$ 7.53	94.94 $\pm$ 3.58	89.01 $\pm$ 3.96
DINOv3	Top Tuning	24.9M	2.80M / 11.27%	70.58 $\pm$ 7.08	83.28 $\pm$ 0.88	90.60 $\pm$ 0.58	85.91 $\pm$ 3.24	80.14 $\pm$ 6.06	93.17 $\pm$ 5.17	89.23 $\pm$ 2.49
YOLO26N	Top Tuning	5.5M	5.50M / 100%	70.59 $\pm$ 6.95	82.09 $\pm$ 3.02	89.83 $\pm$ 1.86	85.14 $\pm$ 4.63	76.77 $\pm$ 5.07	95.64 $\pm$ 4.36	88.48 $\pm$ 3.48
DINOv3	LoRA	24.9M	2.85M / 11.44%	53.78 $\pm$ 9.85	79.45 $\pm$ 2.10	88.11 $\pm$ 1.32	73.49 $\pm$ 5.96	61.78 $\pm$ 7.74	92.07 $\pm$ 6.97	80.23 $\pm$ 3.21
DINOv3	Adapter	24.9M	2.85M / 11.45%	55.17 $\pm$ 6.71	80.28 $\pm$ 1.96	88.66 $\pm$ 1.28	78.21 $\pm$ 3.72	69.49 $\pm$ 3.97	89.50 $\pm$ 3.60	82.85 $\pm$ 2.94
DINOv3	Vis. Prompt	24.9M	2.80M / 11.28%	52.95 $\pm$ 14.47	77.12 $\pm$ 6.43	86.57 $\pm$ 4.14	69.42 $\pm$ 14.54	57.63 $\pm$ 17.46	91.40 $\pm$ 5.82	78.63 $\pm$ 8.08
DINOv3 Classification	Adapter	22.2M	0.90% (199.2K)	-	-	-	90.07 $\pm$ 5.07	89.64 $\pm$ 4.83	90.61 $\pm$ 5.93	91.13 $\pm$ 4.67

method, while foundation models as the backbone are sufficiently good for this task. Based on the detection table and comparing the results of DINO-YOLO with YOLO variants, it becomes visible that the YOLO small model, with comparable parameters to DINO-YOLO, has the same performance, while the number of trainable parameters on DINO-YOLO is much less than that of YOLO small. Another important observation is that during the classification task, the adapters are the best models, but for the detection model, the top tuning reached the best result, this behavior caused due to domain shift between the DINO and YOLO neck; therefore, to mitigate this problem, it would be better to add warm-up layers that first try to tune FPN layer while the neck and backbone are freezes and then gradually the layers unfreez.

In conclusion, foundation models significantly enhance downstream performance while requiring fewer trainable parameters, resulting in faster training and more robust inference.

References

- [1] Paolo Didier Alfano, Vito Paolo Pastore, Lorenzo Rosasco, and Francesca Odone. Top-tuning: A study on transfer learning for an efficient alternative to fine tuning for image classification with fast kernel methods. *Image and Vision Computing*, 142:104894, 2024. 2
- [2] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3, 2022. 2
- [3] Yanghao Li, Hanzi Mao, Ross Girshick, and Kaiming He. Exploring plain vision transformer backbones for object detection. In *European conference on computer vision*, pages 280–296. Springer, 2022. 2
- [4] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation forest. In *2008 Eighth IEEE International Conference on Data Mining*, pages 413–422. IEEE, 2008. 2
- [5] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. 3
- [6] Bernhard Schölkopf, John C. Platt, John Shawe-Taylor, Alex J. Smola, and Robert C. Williamson. Estimating the support of a high-dimensional distribution. *Neural Computation*, 13(7):1443–1471, 2001. 2
- [7] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3, 2025. 1
- [8] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *International journal of computer vision*, 130(9):2337–2348, 2022. 2